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Impact of Dean Vortices on the Integrity Testing of a Continuous Viral Inactivation Reactor

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We propose a standard protocol for integrity testing the residence-time distribution (RTD) in a "Jig in a Box" design (JIB)—a previously described tortuous-path, tubular, low-pH, continuous viral inactivation reactor, ensuring that biopharmaceutical products will be incubated for the required minimum residence time, t_{min}. t_{min} is the time by which just 0.001% of the total product containing virus has exited the incubation chamber (i.e., $t_{0.00001}$). This $t_{0.00001}$ is selected to ensure a >4-log reduction in viral load. As current tracers and in-line analytical technologies may not be able to detect tracers at the 0.001% level, an alternative approach is required. The authors describe a method for deriving t_{min} from $t_{0.005}$ (i.e., the time at which 0.5% of the product has emerged from the reactor outlet) and an experimentally confirmed offset value, $t_{offset} = t_{0.005} - t_{0.00001}$. The authors also evaluate tracer candidates—including 100-nm-diameter gold nanoparticles, dextrose, monoclonal antibody, and riboflavin—for pre-process acceptability and the effects of viscosity, molecular diffusion coefficient, and particle size. The authors show that a JIB will yield t_{min} and RTDs that are nearly identical for multiple tracers due to Dean vortex induced mixing. Results indicate that almost any small-molecule tracer that is generally recognized as safe can be used in pre-use integrity testing of a continuous viral inactivation reactor under the Deans values (De) of 119–595.

1. Introduction

The transition from the current batch mode of processing biopharmaceuticals to a continuous or integrated process promises to greatly improve the way biologics are manufactured by reducing capital and operational costs, minimizing equipment footprint, increasing productivity, and enhancing flexibility.[1] The low-pH viral inactivation step is one unit operation that could benefit from conversion to continuous processing. Compared to batch systems, continuous viral inactivation (CVI) is less complex (with fewer opportunities for deviations), reduces processing time, and requires less chasing solution to remove carry-over product from cycle to cycle. In addition, a CVI unit operation provides a path to a truly continuous end-to-end process.[2]

One of the key steps in developing a new unit operation is the creation of an integrity test to ensure that the system has been fabricated without defects and correctly assembled in the manufacturing setting. This integrity test ensures that, once the critical process parameters are met, the process will reliably achieve the required viral clearance and produce consistent product quality. For low-pH viral inactivation, the critical process parameters include pH, temperature, buffer species, protein concentration, and incubation time.[3,4] In a continuous space, incubation time is particularly challenging to understand (due to dispersion effects), establish, and verify. This paper focuses on establishing an incubation-time integrity test in a CVI tubular reactor.

Low-pH viral inactivation is currently carried out in batch mode by reducing the pH of the product (typically to a level below pH 3.8) and incubating in a continuously stirred tank reactor (CSTR) for some minimum hold time, t_{batch} (a function of the critical process parameters).[4,5] This hold time must be long enough to ensure more than 4 logs of viral reduction (LRV) of enveloped viruses. In manufacturing practice, conservative hold times are often 30 min or more.[4,6]

In the CVI process proposed by Orozco et al.[2] and Parker et al.[7] a stream of product and low pH buffer are mixed with a static mixer prior to entering a tortuous-path tubular chamber (called a "Jig in a Box," or "JIB") with the volume necessary to incubate the product with potential virus for the required hold time at the design process flow rate. To ensure proper incubation, process developers must define the minimum residence time (t_{min})—the interval between the sample's entrance into the reactor and the time the first fluid element containing potential virus appears at the outlet. This t_{min} must be equivalent to the inactivation time used in a batch process, t_{batch}, that ensures ≥4 LRV. Here, we define t_{min} as $t_{0.00001}$ or the time at which no more than 0.001% of the total product with virus stream has exited the reactor. This occurs when the break-through function $F(t_{min}) = 10^{-5}$.

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In a fully integrated or truly continuous process, multiple unit operations are connected, without pooling product between operations. Biomanufacturing processes rely heavily on disposable equipment that will not support the pressures needed to achieve the high Reynolds numbers necessary for turbulent or plug flow. The hold time in the ideal plug flow regime, t_{plug}, is defined as the ideal tubular reactor volume, $V_{idealreactor}$, divided by the process flow rate, Q (Equation 1).

$$t_{plug} = \frac{V_{idealreactor}}{Q} \tag{1}$$

In a continuous process one cannot assume that t_{plug} is the t_{batch} that is used for batch processing. As integrated or connected biomanufacturing process must operate in the laminar-flow regime, there will be some dispersion of the product stream as it flows through the tubular reactor. Because of this dispersion, a significant portion of the product and potential viral contaminants will exit the reactor earlier than t_{plug} and might not be fully inactivated because it is underincubated (**Figure 1**A,B). Therefore, an integrity test that measures the actual t_{min} (which is always less than t_{plug}) and confirms that it is equal to t_{batch} is needed.

It is difficult to directly measure t_{min} at the low level of $F(t_{min}) = 10^{-5}$ using currently available in-line analytical technology. We propose instead to measure the minimum time that one can accurately detect the tracer concentration with in-line analytical technology, and to correlate that time with t_{min} using the break-through function F-curve. $F(t) = 0.005$ offered the lowest concentration measurable in the noisiest conditions investigated in this study. We refer to

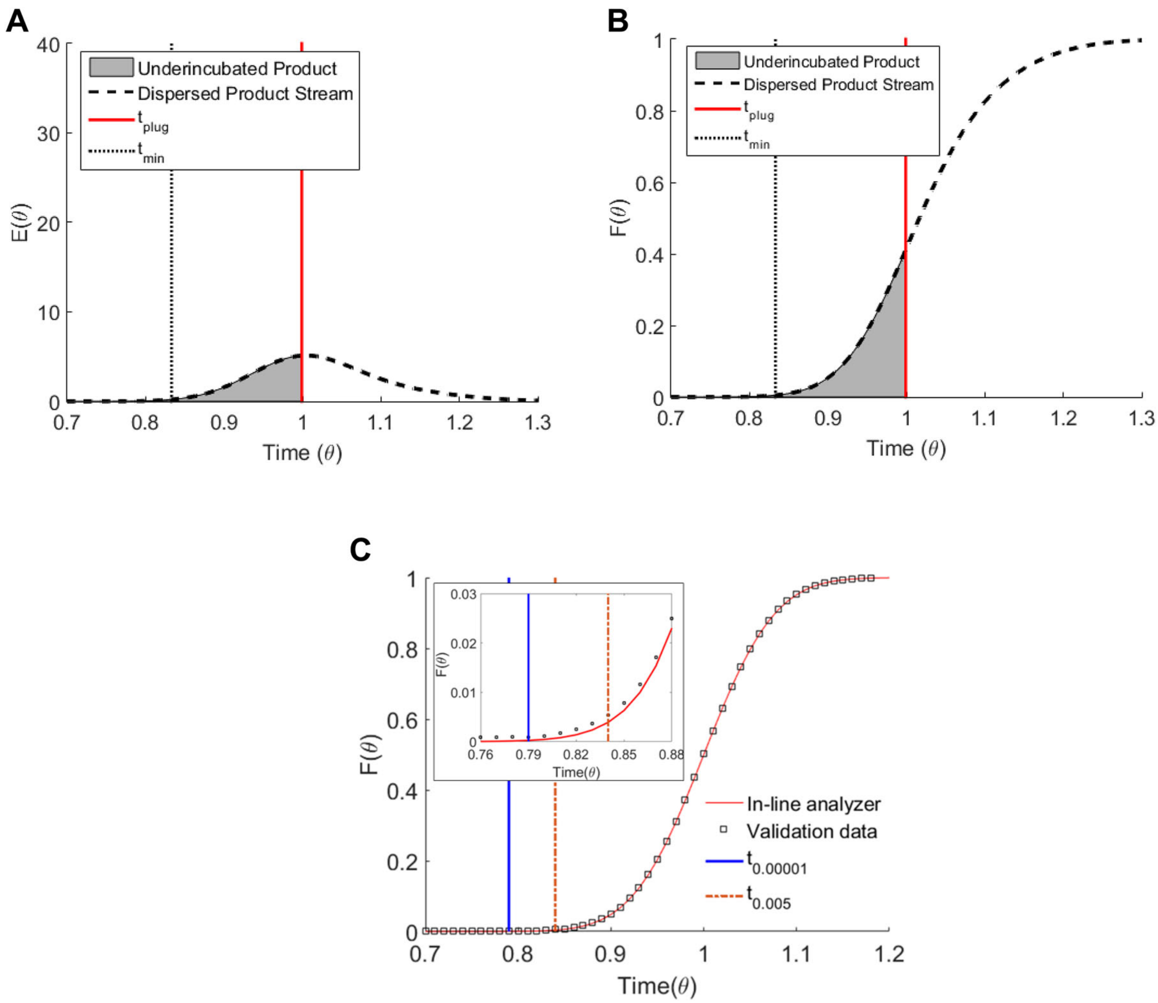

Figure 1. Theoretical residence time distribution of a protein solution. A) The theoretical E-curve gives normalized concentration of the product stream with time. B) The F-curve is the integral of the E-curve and is the percentage of tracer that has exited the reactor. There is dispersion of the product stream, which causes a portion of the product to exit the tubular reactor prior to t_{plug}; this product would be underincubated if t_{plug} is assumed. We thus propose to use t_{min} to determine the effective inactivation time. C) Theoretical Integrity Test of the Continuous Viral Inactivation in a Tubular Reactor. In a manufacturing setting, if all process parameters are fixed, then the inline analyzer data should line up with the validation data. If these data line up, then $t_{0.00001}$, obtained from the validation data, should be equivalent for both data sets and $t_{0.005}$ of the tracer solution can be used to estimate $t_{0.00001}$. The inset is a zoomed in version of the area between $t_{0.005}$ and $t_{0.00001}$, where we show a negligible fraction of underincubated material between the two time points.

the time at which $F(t) = 0.005$, as $t_{0.005}$, which will be larger than $t_{0.00001}$ ($= t_{\min}$) in all cases.

The tubular CVI reactor and the associated integrity test might be validated by spiking a viral contaminant into a scaled-down reactor that mimics the full-scale process as closely as possible, and subsequently measuring the residence time distribution (RTD), LRV, and $t_{\min}$.[6,8] Obviously, however, this approach works only if the small- and at-scale values for $t_{\min}$ and the spread of the curve can be made to match. This poses great challenges, is not practical, and has not been demonstrated.

Another option is to perform small-scale batch experiments to determine the t_{batch} that will provide LRV ≥ 4 under worst-case operating conditions. Once t_{batch} is established, the tubular reactor can be sized based on the process flow rate and a design-specific "non-ideality factor," $\in$, that describes the ratio by which the particular device's design volume must exceed the ideal plug-flow volume (see Equation (1)). For the tortuous-path tubular reactor studied here (the JIB), $\in = 1.05$ for a $t_{\text{batch}} = 60$ min. This means that the actual reactor volume must be about 5% larger than the ideal plug-flow volume.[2] Thereafter, validation experiments of the RTD with live viruses or phages can be conducted in a properly sized at-scale tubular reactor using the desired process flow rate. The advantage of performing the validation at scale is that the data gathered will be the most representative of the conditions in a manufacturing setting. The data can serve as a benchmark to compare against RTD data generated right before or during processing using an in-line analyzer (Figure 1C). When the CVI is being set up for commercial operation, an integrity test will be needed to ensure that the given chamber volume and process flow rate yield a residence time distribution (RTD) and minimum residence time, $t_{\min} = t_{0.00001}$, that match those obtained during the validation studies. This integrity test requires a tracer with flow characteristics that duplicate those of potential viral contaminants in the product stream. If we select a tracer with an RTD matching that of the phage or viral contaminant, we can use $t_{0.005}$ to estimate $t_{0.00001}$ (Figure 1C). A zoomed in version appears in the inset of Figure 1C, where we show that there should be a very small fraction of underincubated product coming out before $t_{0.00001}$.

In selecting candidate tracers for the integrity test, we set three main requirements: First, the tracer's distribution must mimic the distribution of potential viral contaminants in the product stream. A minimum residence time and the variance of the curve can be used as a metrics to characterize the distribution. Second, the tracer must be measurable at concentrations low enough to permit identification of the first tracer-containing fluid elements to emerge from the reactor outlet. Finally, the tracer must be generally recognized as safe (GRAS) and easy to clear from the system, to allow for pre-use integrity testing.

In this work, we evaluated two tracers that fit this description (riboflavin and dextrose) for use in an integrity test, and compare their performance with two other tracers, 100-nm-diameter gold particles and monoclonal antibody. We also provide considerations and recommendations for how the integrity test could best be implemented in a manufacturing setting.

2. Experimental Section

The RTD was measured by introducing a tracer pulse at the inlet of the tubular reactor, chasing the pulse with water, and measuring tracer concentration at the outlet. Four tracers were screened in this study: riboflavin, dextrose, a gold nanoparticle suspension, and a monoclonal antibody (mAb) solution. The tubular "JIB" reactor (described in Orozco et al.) was fully primed with water prior to tracer injection. The tracer (13 mL pulse) and water from the chase line were pushed into the reactor through a four-port, two-way valve with an inner diameter of 6 mm (GE Healthcare Bio-Sciences, PA). Introduction of the tracer and flow of water through the tubular reactor were controlled by two peristaltic pumps (Watson–Marlow 520S IP31, Watson–Marlow, United Kingdom). Peristaltic pumps were selected because they are easy to use and have very little impact on dispersion.[9] Gold nanoparticle suspensions were manually injected into the tubular reactor with a plastic syringe.

Tracer concentrations were measured using both in-line and offline methods. In-line measurements were made with a UV spectrometer (Flow VPE, C Technologies, NJ) in-line with a Cary 60 UV–Vis Spectrophotometer (Agilent Technologies, Santa Clara, CA). Fixed-path-length and step-size methods (start path-length 3 mm; step path-length 0.4 mm) were used to analyze residence-time distributions for all tracers except the mAb solution, for which a variable-path-length method was used to compensate for the narrow linear range of the protein. Offline measurements were done with a total organic carbon (TOC) analyzer (Sievers M9 Laboratory, GE Analytical Instruments, Boulder Colorado). Dilutions of TOC samples were performed to measure concentrations at the peak of the residence time distribution. The tubular JIB reactors used for these experiments are 3D-printed (ProX 800 SLA Technology, 3D Systems, Rock Hill South Carolina) prototypes described in Orozco et al.[2] and Parker et al.[7] The reactors were connected in series via 3-mL sanitary connections to scale up volume from one reactor (530 mL) to six reactors (3.18 L total volume).

2.1. Experimental Setup

A pulse of tracer is introduced into the tubular reactor (JIB). The valve is then switched and the pulse is chased with water. Concentration is measured at the outlet of the reactor with an inline UV analyzer. For TOC experiments, the UV analyzer was removed and samples were taken directly at the outlet of the JIB and measured off-line. The flow rates tested were 50-250 mL min^{-1}, which corresponds to Reynolds numbers (Re) of 167–835 and Dean values of 119–595.

As mentioned, this study used four different tracers (**Table 1**). Gold nanoparticle suspensions, anhydrous dextrose, and riboflavin were all purchased from Sigma–Aldrich. Dextrose (357 g L^{-1}) and riboflavin (50 mg L^{-1}) were both dissolved in water for injection. The mAb solution (60 g L^{-1}) consisted of a 150 kDa molecule dissolved in a solution containing sodium phosphate, Tween 20, and trehalose. Dynamic viscosity was measured using a Gilmont falling-ball viscometer (Cole-Parmer, Vernon Hills, IL).

The un-normalized residence time distribution, $C(t)$, was normalized to yield the exit age distribution, $E(t)$, also known as the RTD (Equation (2)). $E(t)$ is cumulatively integrated to compute the break-through curve, $F(t)$ (Equation (3)). The F-curve gives the fraction of product that has exited the reactor at time t. The plug flow time, t_{plug}, was used to convert time to dimensionless time, θ. Here, L is the path-length of the tubular reactor, and u is the linear velocity.

Table 1. Information on tracers used in this work.

Tracer	Conc. ($g\,L^{-1}$)	TOC[a] (parts per million)	λ_{max} (nm)	Diffusion coefficient[b] ($cm^2\,s^{-1}$)	Viscosity[a] (mPa · s)	Particle diameter[c] (nm)
Au nanoparticle (100 nm)	3.84×10^{-2}	N/A	570	4.29×10^{-8}	1.00	100
Dextrose	3.57×10^{2}	1.43×10^{5}	N/A	2.19×10^{-6}	2.58	0.76
mAb solution	6.00×10^{1}	6.54×10^{4}	280	1.66×10^{-7}	2.58	10.0
Riboflavin	5.00×10^{-2}	2.71×10^{1}	372	3.70×10^{-6}	1.00	1.16

[a] Values were measured; [b] Values were calculated using the Stokes-Einstein equation and known values for the hydrodynamic diameter.; [c] The manufacturer's value was used for the gold nanoparticles. All other values for the hydrodynamic diameter (i.e., particle diameter) were taken from literature sources.[10]

$$E(t) = \frac{C(t)}{\int C(t)dt}; E(\theta) = E\left(\frac{t}{t_{plug}}\right) \cdot t_{plug} \quad (2)$$

$$F(t) = \int_0^t E(t)dt; F(\theta) = F\left(\frac{t}{t_{plug}}\right) \quad (3)$$

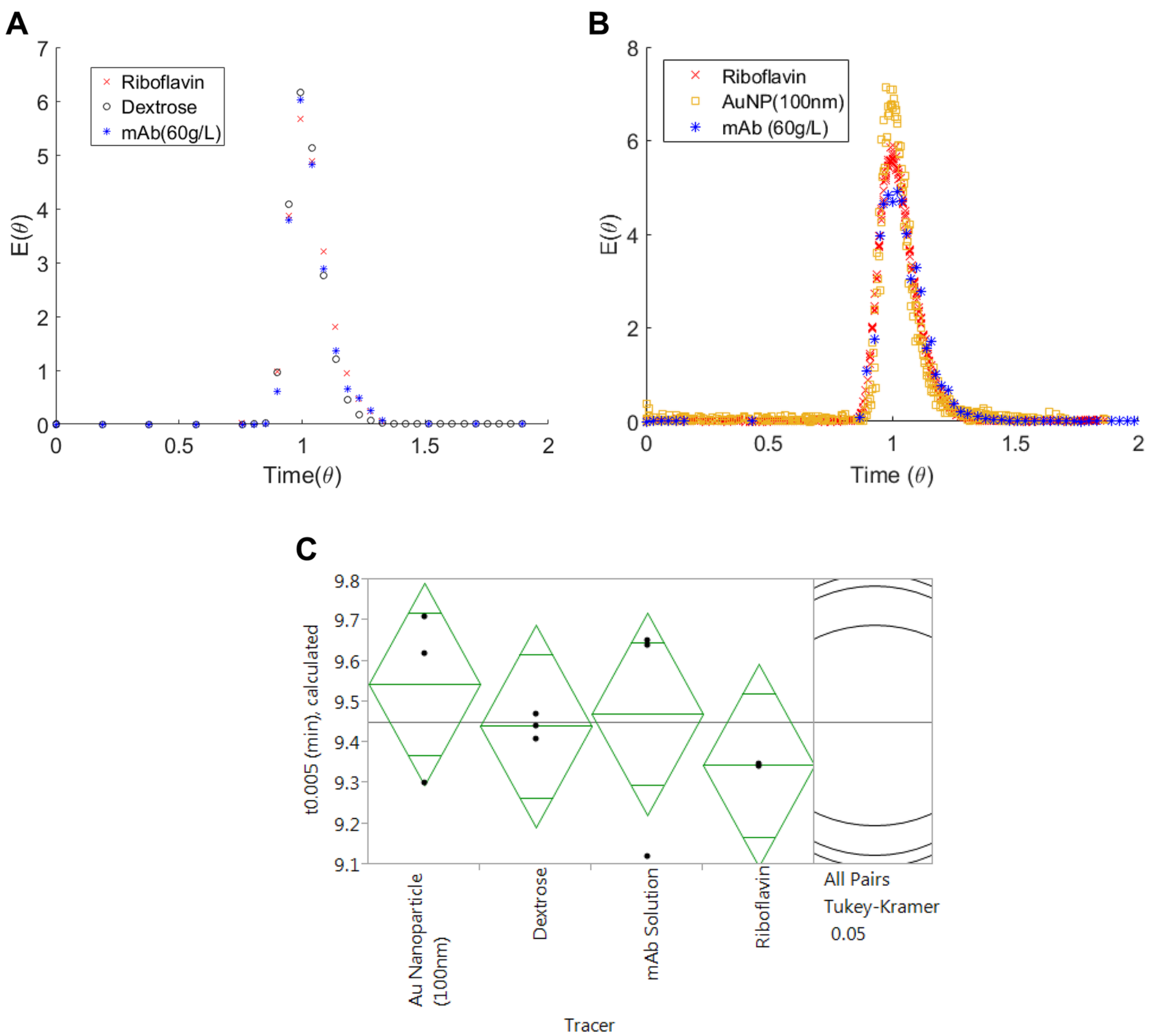

Figure 2. Residence time distribution in a 530 mL reactor for all tracers, measured with TOC (A) and UV Spectroscopy (B). The TOC distributions for riboflavin, dextrose, and mAb match very closely. UV measurements show a marked increase in data noise. The riboflavin and the AuNP residence time distributions nonetheless also closely match that of the mAb solution. C) The analysis of variance (ANOVA) and Tukey tests tell us that all the $t_{0.005}$ values gathered under all different conditions are not significantly different. The values are summarized in **Table 2**.

3. Results and Discussion

3.1. Tracer Selection

3.1.1. Dextrose

Dextrose was initially chosen because it is highly soluble in water. This high solubility allows for TOC measurements at concentrations that are several orders of magnitude lower than the stock concentration of $3.57 \times 10^2\,\mathrm{g\,L^{-1}}$ (Table 1). The ability to measure dextrose at such low concentrations makes it ideal for identifying the first tracer-containing fluid element at the reactor outlet. Additionally, dextrose is a GRAS molecule, with no apparent toxicity.[11] Dextrose is one of many sugars commonly found in mAb formulations.[12] One major problem with dextrose, however, is that many TOC analyzers require 4 min or more to analyze each sample; this means that dextrose must be measured offline.[13] In-line measurements are preferred: they reduce manual sample-handling, which minimizes error, and reduce processing times, allowing real-time monitoring of the product stream as it moves through the reactor. The benefits of real-time data translate to a more reproducible process and consistent quality of the final product.[14]

3.1.2. Riboflavin

To circumvent off-line dextrose analysis, we explored the option of using riboflavin as a tracer. Riboflavin, or vitamin B_2, is an analog of coenzymes flavin adenine dinucleotide (FAD) and flavin mononucleotide (FMN) that bind to flavoproteins and catalyze various reactions in the body.[15] Riboflavin is water-soluble and fluorescent. Riboflavin is also a GRAS molecule and has no apparent cytotoxicity, even when consumed in quantities several times the recommended daily allowance.[16] We selected riboflavin primarily for its fluorescent properties; a fluorescent tracer allows for near-continuous (frequency $= 6\,\mathrm{s^{-1}}$) in-line measurement of the residence-time distribution.

The convenience and high temporal resolution of inline UV measurements come at a cost in accuracy: UV measurements of riboflavin were much noisier than TOC measurements of dextrose. This is because riboflavin has low solubility in water, which allows only a low stock concentration, some four orders of magnitude more dilute than dextrose, Table 1. A low stock concentration reduces resolution, making it difficult to measure the RTD in the tailing regions. In other words, a lower the initial tracer concentration results in a higher limit of detection.

3.1.3. Other Tracers

While dextrose and riboflavin are both GRAS molecules and easy to clear from the tubular reactor, they do not have diffusion coefficients equivalent to monoclonal antibodies or to the viral particles that may contaminate the product stream. We therefore also studied a $60\,\mathrm{g\,L^{-1}}$ mAb solution (duplicating the highest concentration one might expect to elute from the affinity chromatography column) and a 100-nm gold nanoparticle (AuNP) suspension (representing potential contaminating viral particles). AuNPs were primarily chosen for their size and uniformity. According to the Stokes-Einstein equation, a particle's diffusion coefficient is a direct function of its size in cases where the Reynolds number is low.[17] Many of the viruses used as models in previous clearance studies tend to have a characteristic length ranging from 50 to 200 nm.[18] Studying a tracer similar in size to the viruses that will be used in the validation studies allowed us to understand what effect large particles with low diffusion coefficients will have on the RTD. AuNPs, however, are not GRAS and cannot be recommended for use in integrity testing before processing: developers would have to demonstrate that they were able to completely remove them from the process. This leads to unnecessary complications.

3.2. Conservation of the Minimum Residence Time for Molecules of Different Sizes

Initially, we were concerned that differences in the diffusion coefficients and/or different molecular sizes of virus surrogate tracers (such as dextrose or riboflavin) would result in different values of $t_{0.005}$ under the same operating conditions. However, the data we collected shows that this is not the case. Pulse tracer experiments showed that $t_{0.005}$ is conserved for the four types of tracers tested (**Figure 2**A,B and **Table 2**).

Earlier experiments showed that increasing the number of bends in the reactor design creates Dean vortices in the flow; these increase radial mixing and reduce axial product-stream dispersion.[2] The synergy of previously published data and the data in Figure 2 suggests that the $t_{0.005}$ in JIB-like devices should be approximately the same for every tracer tested under the same conditions, and any GRAS tracer that is easy to clear from the system can be used in an integrity test of this type of tubular reactor.

3.3. Conservation of Minimum Residence Time for Relevant Viscosities

One of the major early questions associated with the various tracers selected is quantifying the impact of having pulse injections that have different dynamic viscosities and densities compared to the water chase. To better understand the behavior of variable viscosity, pulse experiments were conducted with

Table 2. $t_{0.005}$ and σ^2 for all conditions tested.

530 mL reactor	
Tracer	$t_{0.005}$ (min), calculated
Au nanoparticle (100 nm)	9.62 ± 0.28
Dextrose	9.44 ± 0.03
mAb solution	9.23 ± 0.06
Riboflavin	9.32 ± 0.08

All values were taken from TOC data except for Au nanoparticle data, which was taken from the UV method of detection.

mAb solutions ranging in concentration from 10 to 60 g L^{-1} and ranging in viscosity from 1.1 to 3.0 cP (**Figure 3**A). The results (Figure 3B) show that the concentration of protein solution entering the tubular reactor has only a minor effect on the residence time distribution at the exit, and the observed shape of the RTD curve changed negligibly with protein concentration (Figure 3B). The distributions matched closely because peak-spreading is influenced mostly by the geometry of the tubular reactor and the flow rate, which do not vary with protein concentration (i.e., viscosity). As we will show, the geometry of the JIB tubular reactor used in this study causes the RTDs of drastically different tracers to converge. This result means that slight variations in tracer concentration will have a negligible effect on the spread of the RTD profile (σ^2) and $t_{0.005}$ at Dean numbers studied.

3.4. Mechanism for the Conservation of Minimum Residence Time and Modeling Radial Mixing in a Curved Pipe

Axial dispersion of product flowing in the laminar regime can result in some product exiting the tubular reactor earlier than t_{plug}. This process is best described by the dispersed plug flow model or axial dispersion model (A.D. Model) (Equation (4) and (9)), where u is the linear velocity, z is axial position, and D_{a} is the apparent diffusion coefficient).[19,20]

$$\frac{\partial C}{\partial t} + u\frac{\partial C}{\partial z} = D_a \frac{\partial^2 C}{\partial z^2} \tag{4}$$

This model assumes that a Fickian diffusion process, characterized by axial dispersion D_a, is superimposed on convective transport of the product stream.[19] Under laminar flow conditions, the front of the product stream will move axially down the tube at a set velocity, u, and assume the parabolic shape characteristic of Poiseuille flow. Parabolic shearing of the product stream increases radial concentration gradients, which leads to increased axial spreading of the product stream.[21,22] This process is known as Taylor dispersion (Equation (5a) and (5b), where d_{t} is the tube diameter and D_{m} is the molecular diffusion coefficient). The Taylor spreading for a straight pipe can be approximated by an effective diffusion term, D_{taylor}, at steady state (Equation (5a) and (5b)).

$$D_{\mathrm{a}} = D_{\mathrm{m}} + D_{\mathrm{taylor}} \approx D_{\mathrm{taylor}} \tag{5a}$$

$$D_{\mathrm{a}} = D_{\mathrm{m}} + \frac{u^2 d_t^2}{192 \cdot D_{\mathrm{m}}} \approx \frac{1}{192} \cdot \frac{u^2 d_t^2}{D_{\mathrm{m}}} \tag{5b}$$

Equation (5b) shows that the rate of spreading due to Poiseuille flow, D_{a}, in a straight pipe becomes inversely proportional to D_{m} when D_{m} is very small. As this is normally the case, Taylor diffusion, D_{taylor}, typically dominates for velocities, u, expected in bioprocessing. If the molecular diffusion coefficient is small, the probability increases that the particles remain near the wall. However, if D_{m} is high, compared to the time constant for radial diffusion (Equation (6)), then the radial diffusion component mixes the particles between regions of high and low velocity in the tube, resulting in a narrow RTD. Regardless of the D_{m} magnitude, the product stream approaches a normal Gaussian distribution at the time limit τ (Equation (6), where τ is the time needed for particles in the product stream to sample all radial positions and r_{t} is the radius of the tube).[21,23]

$$t_{\mathrm{plug}} \gg \tau = \frac{r_{\mathrm{t}}^2}{D_{\mathrm{m}}} \tag{6}$$

Equation (5b) presumes that the product stream is moving through a straight tube, in which the only radial transport mechanism is molecular diffusion. In the tortuous-path viral inactivation reactor evaluated here, however, a series of bends deflect the flow path, producing centripetal forces that pull the fluid radially outward from the center of curvature. The

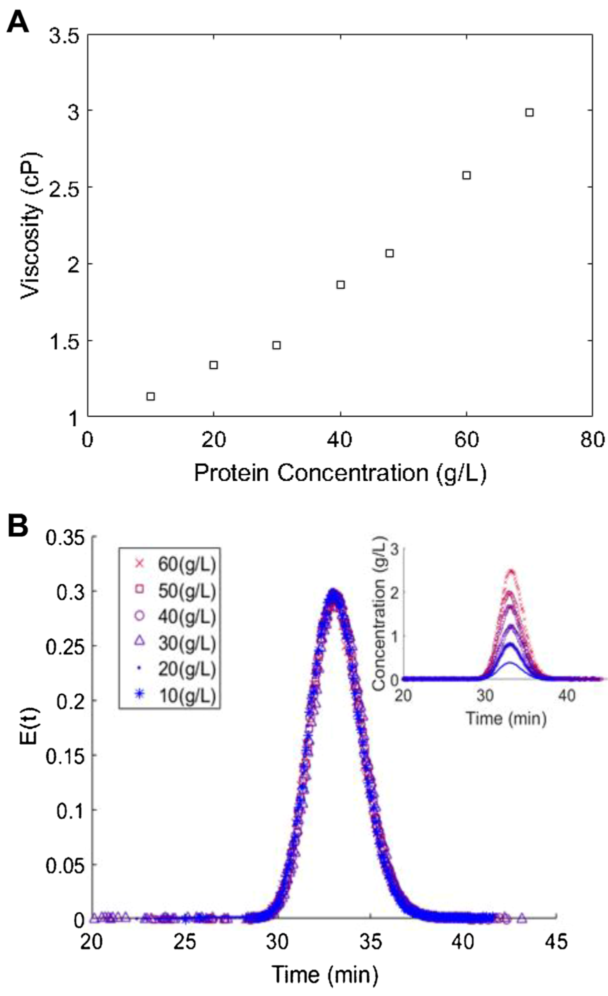

Figure 3. A) Dynamic viscosity for mAb solutions at different concentrations. B) Pulse tracer experiments with mAb solutions at different concentrations at Re = 334 and De = 238. Dispersion is a linear process predominantly controlled by convective forces, which are independent of protein concentration in the range of Dean values studied. Therefore, the normalized RTDs agree closely for all protein concentrations lower than 60 g L^{-1}.

acceleration creates a pressure gradient that pushes the fluid closer to the flow path's center of curvature. The interaction between these two radial forces produces a pair of Dean's vortices in the transverse plain of the tube.[24] These Dean's vortices increase radial mixing, resulting in anisotropic dispersion of the tracer. The intensity of the mixing is quantified by the Dean number (Equation (7)), which is the Reynolds number, Re, scaled by r_c, the ratio of the radii of curvature of the tube and the flow path.[25]

$$De = \left(\frac{d_t}{2r_c}\right)^{1/2} Re \tag{7}$$

We account for the added radial dispersion of the Dean's vortices by replacing the axial component of D_m (Equation (5a)) with the effective radial component Dr (Equation (8a), where Dr is the effective radial dispersion coefficient and it is the sum of the radial component of D_m and D_v). D_v is the vortex dispersion coefficient (Equation (8b)). D_V reflects the increase in convective radial transport caused by Dean vortices. We argue that the increased radial convection caused by Dean vortices can be approximated by a Fickian transport mechanism after the particles have sampled many radial positions or the radial concentration gradient approaches a steady state over the long time limit, τ (Equation (6)).

$$D_a = D_m + \frac{1}{192}\cdot\frac{u^2 d_t^2}{D_r} \approx \frac{1}{192}\cdot\frac{u^2 d_t^2}{D_r} \tag{8a}$$

$$D_r = D_m + D_v \approx D_v \tag{8b}$$

In this model, Dr is much larger than D_m, which results in a pulse that approaches the normal Gaussian distribution much faster than in a straight tube and has a narrower distribution.

Under the model proposed by Taylor for straight tubes (Equation (5a) and (5b)), we expect to see an inverse relationship between the D_a and D_m (**Figure 4**A). However, our experimental results show that the effective dispersion coefficient in the JIB under the velocities studied, is independent of D_m and it is constant, 27 cm s^{-2}. The effective D_a value for the JIB is several orders of magnitude lower than that predicted by Taylor's model for a straight pipe (compare Figure 4A,B). This is due to the increased mixing caused by the Dean vortices. Using Equation (8a) and (8b), we measured D_v at approximately 5.22×10^{-4} cm s^{-2}, and used D_v to calculate the radial transport time constant, τ_{radial}, at 3.22 min. This indicates that this radial transport

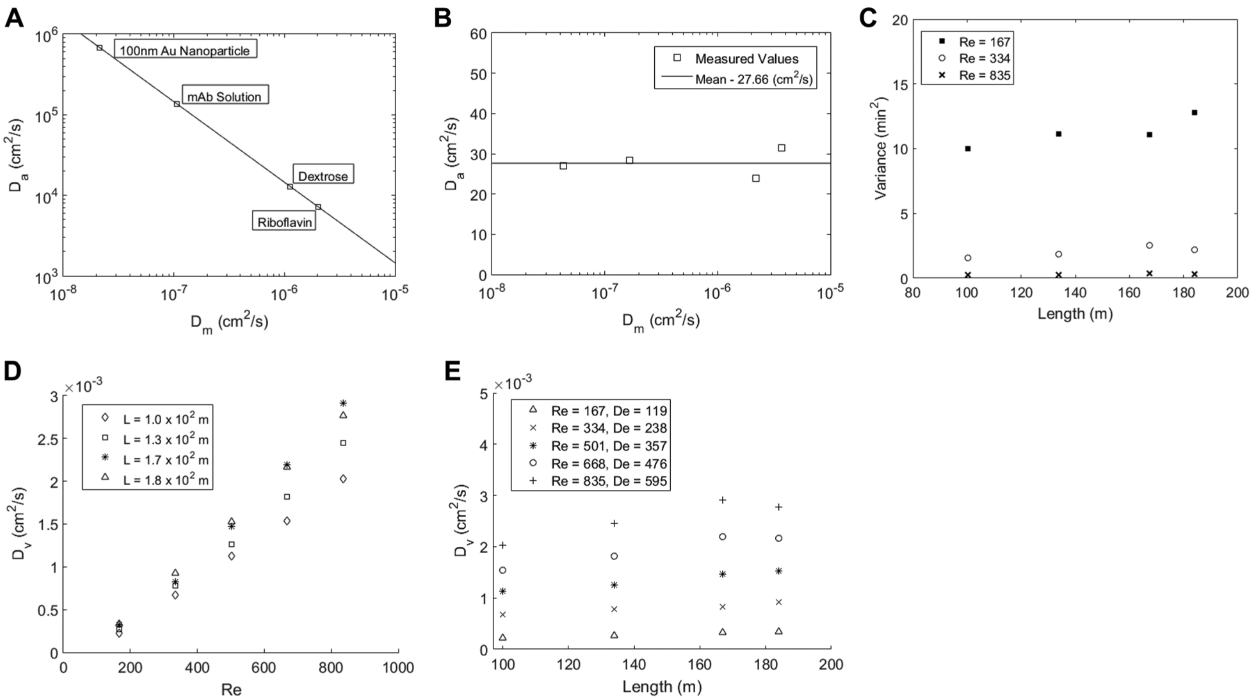

Figure 4. A) The expected apparent dispersion coefficient of each tracer in the case where Taylor dispersion dominates. B) Measured values of the apparent dispersion coefficient show that the values are independent of the molecular diffusion coefficient and much lower in magnitude than the dispersion coefficient predicted by Taylor's model, indicating that convection due to Dean Vortices dominates the system and Equation (8a) is an appropriate model of this system. C) Variance of pulses exiting reactors of different volumes. As the volume of the tubular reactor increases, the variance of the RTD increases linearly. The slope of this relationship approaches zero as the flow rate increases and the system approaches plug flow. The increased flow rate enhances radial mixing, which reduces axial spreading. In the laminar regime, this mixing occurs via Dean vortices as previously described. At higher, turbulent flow rates, however, mixing is facilitated by turbulent mixing or eddy diffusion. D) We expect D_v to increase with Reynolds number, which is shown in the data. E) D_v is independent of length at a fixed Reynolds number. Increasing the Reynolds number will increase Dean vortices and thus increase D_v.

mechanism dominates beyond the radial time constant for the particular De numbers used in this study.

The effect of D_v is orders of magnitude larger than that of D_m for the tracers tested in this study. As D_v is much larger than D_m we would expect to see an apparent diffusion coefficient that is independent of D_m and much lower than that predicted by Taylor's model, and our results show this to be the case. It can be helpful to re-cast the dispersed plug flow model or axial dispersion model (Equation (4)) in dimensionless time (Equation (9)).[19]

$$C(\theta) = \frac{1}{2\sqrt{\pi(D_a/uL)}} \cdot e^{-\frac{(1-\theta)^2}{4\left(\frac{D_a}{uL}\right)}} \tag{9}$$

This equation is analogous to the general equation for a Gaussian distribution. If we use the exponent term from the Gaussian function and compare it with the exponent in Equation (9), we can conclude that variance indeed scales linearly with L. This phenomenon is known to be true for laminar-flow systems.[19] This was shown in Figure 4C, where we also show that the slope of the variance-length line approaches the asymptote of plug flow as the Reynold's number increases.

Also, D_v can be seen as an analogue of the Dean's number, in that it quantifies the strength of the Dean's vortices. As De varies directly with flow rate, we would expect D_v to vary with flow rate in the same way, and we see this experimentally (Figure 4E). De and the strength of Deans vortices, D_v, increase with Re and the linear velocity. We see this, too, in the experimental results (Figure 4D) which show a linear relationship between Re and D_v.

3.5. Measurement of $t_{min} = t_{00001}$ and t_{005}

As suggested earlier, there is an offset value between $t_{0.005}$ and the proposed $t_{min} = t_{0.00001}$ (Figure 1). This offset will increase as the residence time in the reactor increases (e.g., total reactor length must increase for a given flow rate). Earlier, we also stated that there is a direct relationship between variance and length (Figure 4C). This means that, as the required incubation time increases for a given flow rate, the offset interval between $t_{0.005}$

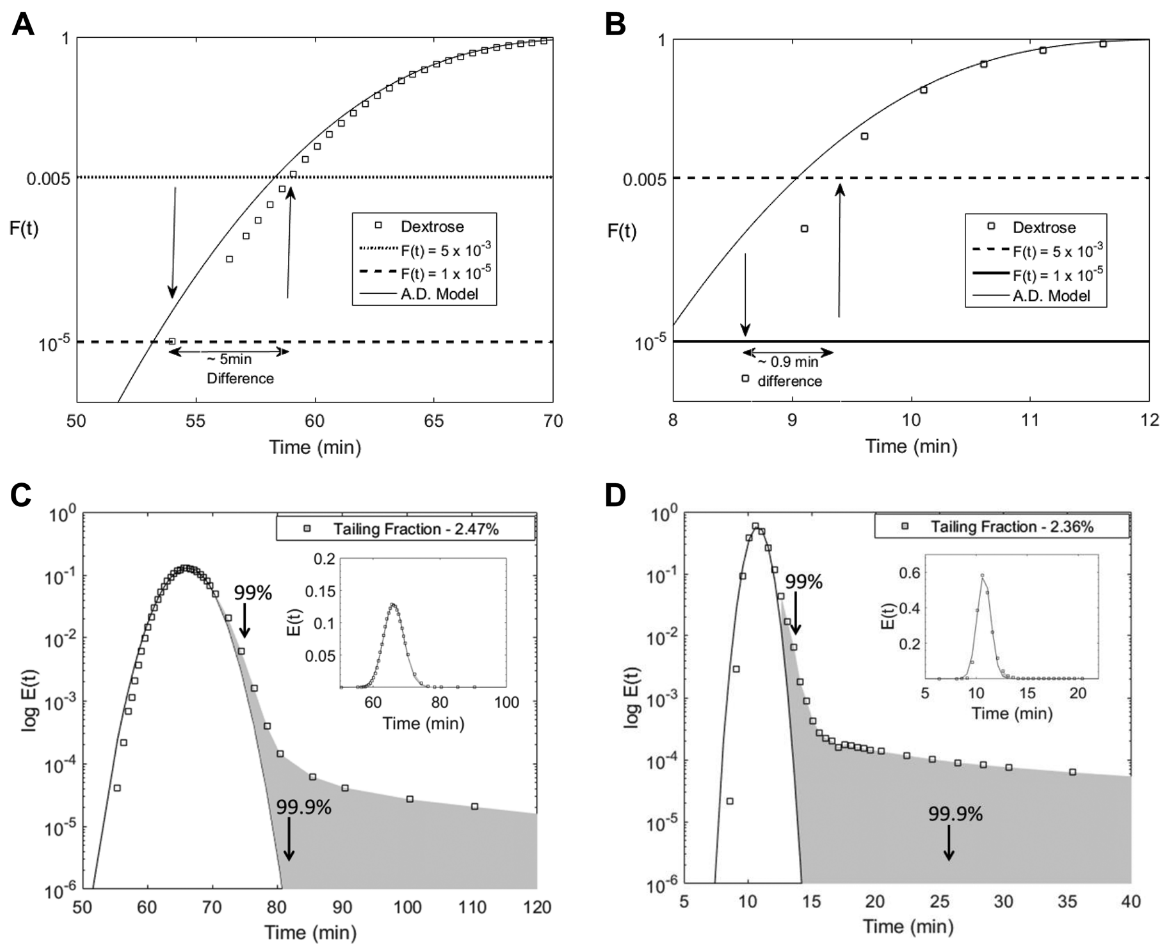

Figure 5. A, B) Offset between $t_{0.00001}$ and $t_{0.005}$ for F curves from dextrose pulse tracer experiments in a (A) 3.18 L chamber and (B) 530 mL chamber. A) Dispersion increased the offset between the $t_{0.00001}$ and $t_{0.005}$ to approximately 5 min. B) The offset was 95 s for the 530 mL chamber. The RTD spreading increase due to length is explained by the axial dispersion model (A.D. Model, Equation (9)), which is conservative at the leading edge compared to empirical data. C, D) Semi-log E-curve showing the amount of dextrose unaccounted for by the axial dispersion model on the tailing edge of the distribution for the (C) 3.18 L and (D) 530 mL chambers.

and $t_{0.00001}$ will also increase. Notice that for the 3.18 L reactor, the offset between $t_{0.005}$ and $t_{0.00001}$ is 5 min. This compares with an offset of about 0.9 min for the 530 mL reactor (**Figure 5**A,B).

To capture this offset, the Axial Dispersion Model (Equation (9)) was employed to compare empirical data for two different sized tubular reactors with dextrose tracer injected at 50 mL min^{-1} (Figure 5A–D). The model follows a good fit for the peak of both conditions and provides a conservative estimate for the value of $F(t_{\min}) = 10^{-5}$. The results provide confidence in the model to obtain conservative estimates of $t_{0.00001}$ if necessary. The tailing end fraction, however, does not differ significantly between different sized tubular reactors. The difference in the total tailing fraction constitutes 2.47% for the 3.18 L reactor and 2.36% for the 513 mL reactor. In any case, if we only care about 99% or 99.9% of the total product, then the fit closely matches the experimental data (Figure 5C,D and 5C,D insets). These results are significant in that we have shown that the Axial Dispersion Model provides a conservative estimate for $t_{0.00001}$.

The change in the offset between $t_{0.005}$ and $t_{0.00001}$ must be taken into account when altering process parameters, like reactor volume. As mentioned in the introduction, the "non-ideality factor," $\in$, tell us how much bigger the tubular reactor has to be in order to meet the $t_{\min}$ requirement. $\in$ is defined as:

$$\in \; = t_{\text{plug}}/\text{t}_{min} \tag{10}$$

In previously published literature, $\in$ was based on a $t_{\min} = t_{0.005}$. In this work, we propose that the non-ideality factor be updated such that the $t_{\min}$ used is $t_{0.00001}$. The change results in 10% more volume needed for the design volume to meet the $t_{0.00001}$ requirement. This means that $\in$ should be 10% greater than it was reported in Orozco et al.[2] For example, instead of $\in \; = 1.05$, it should be $\in = 1.15$ for a $t_{\text{batch}} = 60$ min. This means that the actual JIB reactor volume needs to be 15% bigger than the ideal plug flow volume for a 60-min target incubation.

4. Conclusions

We have laid out the steps that must be completed to create an integrity test for the incubation time in a continuous viral inactivation reactor. First, developers must choose a tracer that is safe and easily cleared from the reactor. Next, they must complete a live-virus validation that ensures that RTDs of viral particles and tracers match. In a tubular reactor with a tortuous path, this should not be a problem because radial mixing will greatly reduce axial dispersion and result in virus and tracer distributions that match closely—that is, the measured values of $t_{0.005}$ and $t_{0.00001}$ should be indistinguishable for both tracer and test virus.

We have presented a model for predicting mixing behavior in a tortuous-flow-path reactor for Dean Numbers of 119–595. This model allows fine-tuning of flow rate and chamber volume based on process requirements. Because inline detection methods, such as UV and conductivity analyses, are far noisier than the offline methods used in this study, it is important to use a higher outlet-fraction threshold like $F(t_{0.005}) = 0.005$. Basing the results at $t_{0.005}$ and using the associated t_{offset} to calculate $t_{0.00001}$ is more realistic and practical than attempting to measure $F(t_{0.00001}) = 10^{-5}$ directly. If the process flow rate and the volume of the reactor are consistent, then the offset time between $t_{0.00001}$ and $t_{0.005}$ will be constant despite changes in protein concentration or viscosity ranging from 1-3cP. Therefore, for conceivable continuous viral inactivation conditions, $t_{0.005}$ can be used to estimate $t_{0.00001}$. Further studies in the validation process will establish acceptability criteria for $t_{0.005}$ and $t_{0.00001}$ based on the hold times required to inactivate viruses and process variability.

Abbreviations

θ, dimensionless time; $\in$, non-ideality factor that indicates how much bigger the design reactor volume used in the process has to be compared to the ideal plug flow volume; σ^2,variance of the residence time distribution; τ, time needed for particles in the product stream to sample all radial positions; $C(t)$, concentration at exit of tubular reactor; D_a, effective dispersion coefficient or axial dispersion coefficient; De, dimensionless deans number; D_m, molecular diffusion coefficient; $E(\theta)$, distribution function for residence times; $F(\theta)$, fraction of material in outflow which has been in system for a time less than theta; JIB, Jig in a Box; L, length of tubular reactor; LRV, log reduction value; Q, flow rate; Re, Dimensionless Reynolds number; r_t, radius of the tubular reactor; RTD, residence time distribution; $t_{0.005}$, minimum residence time measured by in-line analyzer; $t_{0.00001}$, true minimum residence time equivalent to t_{batch}; $t_{0\text{offset}}$, $t_{0.005} - t_{0.00001}$; t_{batch}, incubation time that ensures more than $\geq$4 logs of viral reduction in a batch process; $t_{\min}$, minimum residence time; t_{plug}, plug flow residence time; TOC, total organic carbon; u, linear velocity; V_{reactor}, volume of tubular reactor.

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Conflict of Interest

The authors declare no commercial or financial conflict of interest.

Keywords

axial dispersion, continuous bioprocessing, Dean number, fluid dynamics, integrity test, minimum residence time, viral inactivation

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